

Research Unemployment to GDP

Studentnames and studentnumbers here

2025-06-24

Set-up your environment

```
require(tidyverse)
```

```
## Loading required package: tidyverse
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2    3.5.2      v tibble     3.3.0
```

```
## v lubridate  1.9.4      v tidyr      1.3.1
```

```
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
require(readxl)
```

```
## Loading required package: readxl
```

```
require(rworldmap)
```

```
## Loading required package: rworldmap
```

```
## Loading required package: sp
```

```
## ### Welcome to rworldmap ###
```

```
## For a short introduction type : vignette('rworldmap')
```

```
require(dplyr)
```

```
require(ggplot2)
```

Unemployment in Europe

Chris van Weerdenburg(2827238), Arsema Tewelde(2863982), Alan Peralta Camacho(2810368), Benjamin Mutlu(2846678), Cihan Öztürk(2769294), Israa Shidai(2849708), Jeremy Li(2848651)

Tutorial group number: 3

Tutorial lecturer: Jack Fitzgerald

Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

Unemployment in the European Union (EU) is a significant social issue that affects millions of individuals, families, and communities across all countries within Europe. It refers to the condition where people who are willing and able to work are unable to find suitable employment. This issue impacts not only those directly affected by this dilemma but also the broader economy and social cohesion within Europe. This is relevant because:

High unemployment rates can lead to decreased consumer spending, lower tax revenues, and increased government spending on social welfare programs. This slows down economic growth and can also hinder post-crisis recovery efforts, for example COVID-19.

It also causes long-term unemployment, which can lead to poverty, social isolation, and reduced access to healthcare and education. It often disproportionately affects young people, older workers, migrants, and those with lower educational qualifications, contributing to widening social inequality.

Youth unemployment is particularly pressing in many EU countries, with some regions experiencing rates above 30%. This delays independence, contributes to brain drain (migration of skilled youth), and weakens future labor market prospects.

Unemployment is closely linked to poor mental health outcomes such as depression, anxiety, and a loss of purpose or identity. The longer someone remains unemployed, the more profound the psychological effects. Many studies show that unemployment leads to depressive symptoms but also a lot of stress, because as a person you lack stability and that causes a feeling of fear.

Here are two important sources that highlight the serious consequences of unemployment. The first source comes from an article by the U.S. Department of Health called *Healthy People 2030*. This report states that work is one of the five social factors of health. According to this report, unemployment can actually cause physical conditions such as high blood pressure, but also psychological problems (Employment – Healthy People 2030 | odphp.health.gov, z.d.).

Another source that highlights the social effects of unemployment is a report from *ScienceDirect*, written by Pohlen. This report states that losing your job in the long term leads to a decrease in social integration, which means actively participating in social activities, and that leads to long-lasting depression (Pohlen, 2019).

Persistent unemployment can erode trust in institutions, fuel populism and extremism, and lead to social unrest. Citizens may feel disillusioned with governments and the EU itself, challenging the union's stability and unity.

There have been quite a few studies on the social problem of unemployment, but there is no research that deeply analyzes the relationship between unemployment and economic growth such as GDP. This report focuses on the economic side by looking at how unemployment affects the economy, as well as the social consequences it brings.

Part 2 - Data Sourcing

We have the following datasets:

- **Unemployment by sex, age and duration of unemployment (Eurostat)**
- **Real GDP growth rate - volume (Eurostat)**
- **Population, aged 15 - 74 - EU labour force survey (Eurostat)**

These datasets are published and sourced by Eurostat. Eurostat is the statistical office of the European Union. They source their data through the ESS which is the European Statistical System. These datasets are also collected by each country's own National Statistical Institute for example the data Eurostat collects from the Netherlands is from the CBS (Centraal Bureau Statistiek). This data collection is done according to established EU regulations and standards, ensuring a degree of harmonization across the EU. The national authorities then transmit the verified and analyzed data to Eurostat.

These datasets have some differences but also similarities.

Key differences:

- GDP per capita relates to overall economic performances, while youth unemployment reflects specific labor market outcomes and the labour force refers to the amount of people between age 15 and 74 who are either employed or unemployed.
- GDP includes the whole population; youth unemployment only concerns young people who are economically active and the labour force includes people between the ages 15 and 74.
- GDP is measured in economic value (adjusted for PPS); youth unemployment is a rate (percentage); the labour force is measured in amount of people.

Complementary:

- A country may have high GDP per capita but also high youth unemployment, indicating that economic gains are not evenly distributed or accessible.
- Using our datasets together provides a more comprehensive view of socio-economic well-being, economic performance (GDP) and inclusiveness or opportunity (youth employment).
- This combination can be especially useful for analyzing whether economic growth is benefiting younger generations.
- Usage of these different datasets gives us a broad insight and amount of data that we could use each and every aspect of the data that we sourced, in order to create clear and explicit plots.

These two datasets were a perfect match for our topic because they gave us an in depth look into the correlation between the GDP growth rate and Unemployment rates but also the unemployment rates and sex, age and duration of unemployment these datasets were crucial for us to conclude whether Covid-19 had an impact on unemployment rates. We added in another dataset which provided the data for the total population aged from 15 to 74 usually residing in Member States in private households, except for persons living in collective or institutional households (e.g. hospitals, residential homes, prisons, military barracks, religious institutions, etc.). With use of this data we could calculate the ratio of population and amount of unemployed. Since with the datasets we had used it did not provide this information. That made it quite hard to properly understand how many people were unemployed since some countries have much bigger populations than others.

MetadataDataset informationAdd to 'My datasets'

Selection

The datasets we used were quite complete however some countries the data was not sourced for example: Bosnia and Herzegovina and Lichtenstein. However we did add in a new data set because the other data sets did not give as a good insight about the percentage of the population being unemployed. It just gave us the amount of people that are / were unemployed. This was a slight hiccup since for example Germany has a bigger population than Norway. Other than that were there no complaints about the datasets, and were they very explicit.

2.1 Load in the data

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

```
head(GDP_df[, (1:3)], n = 5)
```

```
## # A tibble: 5 x 3
##   'Data extracted on 04/06/2025 09:25:25 from [ESTAT]' ...2      ...3
##   <chr>                                                <chr>      <chr>
## 1 Dataset:                                           Real GDP growth ra~ <NA>
## 2 Last updated:                                     03/06/2025 23:00    <NA>
## 3 <NA>                                                <NA>      <NA>
## 4 Time frequency                                     <NA>      Annu~
## 5 Unit of measure                                   <NA>      Chai~
```

```
head(Unemploymentlang[, (1:3)], n = 5)
```

```
## # A tibble: 5 x 3
##   'Data extracted on 03/06/2025 10:04:59 from [ESTAT]' ...2      ...3
##   <chr>                                                <chr>      <chr>
## 1 Dataset:                                           Unemployment by se~ <NA>
## 2 Last updated:                                     14/04/2025 23:00    <NA>
## 3 <NA>                                                <NA>      <NA>
## 4 Time frequency                                     <NA>      Annu~
## 5 Unit of measure                                   <NA>      Thou~
```

```
head(Total_Work_population[, (1:3)], n = 5)
```

```
## # A tibble: 5 x 3
##   'Data extracted on 18/06/2025 10:18:37 from [ESTAT]' ...2      ...3
##   <chr>                                                <chr>      <chr>
## 1 Dataset:                                           Population, aged 1~ <NA>
## 2 Last updated:                                     12/06/2025 23:00    <NA>
## 3 <NA>                                                <NA>      <NA>
## 4 Time frequency                                     <NA>      Annu~
## 5 Unit of measure                                   <NA>      Thou~
```

Unemploymentlang: this dataset contains 21 columns and 56 rows. The first 9 rows, row 11 and rows 50 to 56 provide only N/A's, representing headers, notes, or non-country data. All odd-numbered columns, except the first, contain N/A's and observation flags, which in this dataset are represented by the letters 'b' (break in time series), 'd' (definition differs) or 'bd' (both). The even-numbered columns hold unemployment figures (in thousands) for 2015-2024, with '.' indicating unavailable data. The columns names are unclear and uninformative.

GDP: The dataset has 25 columns and 59 rows. The first 7, 9th and last 7 rows consist of N/A's, since they only include headers, notes, titles. All odd-numbered columns are empty (N/A) except for the first. However, they contain observation flags: 'b', 'p' (provision) or 'e' (estimated). Even-numbered columns present the GDP percentage changes from 2013-2024, however with some countries the data is unavailable. This is represented by a '.'. The columns are only numbered and uninformative.

Total_Work_Population: This dataset consists of 50 columns and 46 rows. The first 10, the 12th and the last 7 rows are empty (N/A's) besides the column headers, such as country, sex and age groups. All odd numbered columns, but the first one are empty, meaning they only contain N/A's and observation flags such as 'b', 'd' or 'bd'. Even-numbered columns are about the actual data: number of employed people in thousands for years 1991-2024. Whenever data is unavailable, it is implied with a colon ':'. The columns names are only numbers and tell nothing of importance.

Each dataset complements the other. The first and second dataset show the relationship between a country's economic performance (GDP) and its unemployment rate. The second and third dataset illustrate the unemployed to labour force population ratio. This manner we can see the unemployment rate for each country, proportionally.

2.3 Describe the type of variables included

The three datasets provided contain various types of variables primarily focused on the employment statistics (Unemploymentlang), GDP growth rate (GDP) and labour force population (Total_Work_Population) which relate to socioeconomic status (SES). Obvious variables such as time in years and place in countries are in each dataset.

Eurostat is the source of all three datasets. They do not gather the data themselves. As previously stated, the data is collected by national statistical authorities in EU countries. These authorities evaluate, analyse and report national data to Eurostat (European commission, 2025).

Part 3 - Quantifying

3.1 Data cleaning

We will start with the unemployment dataset. We will remove the first two columns, as they are not relevant for our analysis.

```
head(Unemploymentlang, n = 3)
```

```
## # A tibble: 3 x 11
##   Country '2015' '2016' '2017' '2018' '2019' '2020' '2021' '2022' '2023' '2024'
##   <chr>   <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 Belgium 421.8 389.6 353.7 300.7 273.7 282.4 324.4 293.7 294.2 306.2
## 2 Bulgaria 305.1 247.2 206.9 173.3 131.7 155.5 156.7 127.3 132.4 127.4
## 3 Czechia 268.0 211.4 155.4 121.5 109.0 137.0 150.5 117.6 134.0 138.8
```

This data, as mentioned earlier, contains information about unemployment in Europe. But in the original data, there were many unnecessary rows that were not relevant. For example, the first rows contained metadata like titles and explanations. There were also many empty columns or columns with NA. By cleaning this all up and keeping only the countries and the years 2015 to 2024, it looked much clearer. Also, doing a clear analysis from the dataset was now much easier. Here is a small summary of which functions were used to clean the data:

-The first 13 rows were not relevant and were removed using `Unemploymentlang[14:49, ]`

-Empty columns and unnecessary columns were removed with `Unemploymentlang[, -c(3, 5, 7, 9, 11, 13, 15, 17, 19, 21)]`

-And lastly, the column names were replaced with the correct years using `colnames(...)[2:11] <- as.character(2015:2024)`

```
head(GDP_df, n = 3)
```

```
## # A tibble: 3 x 13
##   Country '2013' '2014' '2015' '2016' '2017' '2018' '2019' '2020' '2021' '2022'
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Belgium    0.3    1.8    1.5    1.2    1.5    1.9    2.4   -4.8    6.2    4.3
## 2 Bulgaria  -0.5    0.9    3.4    3      2.7    2.5    3.8   -3.2    7.8    4
## 3 Czechia    0      2.2    5      2.6    5.2    2.8    3.6   -5.3    4      2.8
## # i 2 more variables: '2023' <dbl>, '2024' <dbl>
```

As mentioned earlier, this data contains the annual growth percentages of the GDP. In the original data, there were a lot of empty columns and also a lot of unnecessary information. There were also NA present. The goal was to focus on the years from 2013 to 2024 and to remove everything outside of that. By cleaning the data this way, only the relevant information remained and the dataset became a bit smaller. In an ideal situation, the data would already have been cleaned properly and focused on 2013–2024. Here is a small summary of which functions were used to clean the data:

-The first 9 rows were removed using `GDP_df <- GDP_df[10:52,]`, and then also the next 3 rows using `GDP_df <- GDP_df[-c(1,2,3), ]`, so that the dataset started directly with the countries

-Empty columns with NA symbols were removed using `colSums(!is.na(...)) > 0` and `[, -c(...)]`

-The column names were then changed to the years from 2013 to 2024 using `colnames(GDP_df)[2:13] <- as.character(2013:2024)`

-All values were also converted to numeric using `lapply()`

```
head(Total_Work_population, n = 3)
```

```
## # A tibble: 3 x 11
##   Country '2015' '2016' '2017' '2018' '2019' '2020' '2021' '2022' '2023' '2024'
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Belgium  8317  8360.  8359  8411.  8453.  8497.  8534.  8596.  8658.  8698.
## 2 Bulgaria 5563. 5510. 5455. 5398. 5070. 5005. 4956. 4901. 4863. 4844.
## 3 Czechia  8201  8174.  8162.  8137.  8130.  8131.  8099.  7924  7764  7880
```

The data set of the total working population contained a lot of years, going all the way back to 1995. But for our project, it was only relevant to have the years from 2015 to 2024. By deleting the other years, it became a lot clearer and more focused on the correct years. The reason why it was chosen to start from 2015 is because from those years on, a lot of data was available and it is also more convenient for the project. In an ideal world, it would have been nicer if all the unnecessary information had already been removed. A few things were done to clean the data; here is a small summary of them:

-The first 12 rows consisted of irrelevant things that were removed using `Total_Work_population[13:39, ]`

-All the years before 2015 were removed with `[, -c(2:41)]`

-Then, all empty intermediate columns with NA were removed with `[, -c(3, 5, 7, 9, 11, 13, 15, 17, 19, 21)]`

-And finally, the column names were renamed with the correct years using `colnames(...)[2:11] <- as.character(2015:2024)`

3.2 Generate necessary variables

Variable 1: Classification by Economic Growth

To compare unemployment rates across Europe, we divided the countries into three equally sized groups based on total GDP growth. This allows us to compare unemployment rates across different levels of economic growth.

```
GDP_df <- GDP_df %>%
  rowwise() %>%
  mutate(
    GDP_growth_total = round((prod(1 + c_across(`2015`:`2024`)) / 100, na.rm = TRUE) - 1) * 100, 2)
  ) %>%
  ungroup()

merged_df <- inner_join(Total_Work_population, Unemploymentlang, by = "Country")

unemployment_percent_df <- merged_df["Country"]

newx_df <- inner_join(
  inner_join(GDP_df, Unemploymentlang, by = "Country"),
  unemployment_percent_df,
  by = "Country"
)

breaks27 <- quantile(
  newx_df$GDP_growth_total,
  probs = c(0, 1/3, 2/3, 1),
  na.rm = TRUE
)

newxx_df <- newx_df %>%
  mutate(
    EconClass = cut(
      GDP_growth_total,
      breaks = breaks27,
      labels = c("Low growth", "Middle growth", "High growth"),
      include.lowest = TRUE
    )
  )
```

First, we used all available data to create a new column representing the total GDP growth. Based on this, we divided the countries into three equally sized groups. This allows us to compare unemployment rates relative to each country's level of economic growth

```
head(newxx_df[, c(1,25)], n = 5)
```

```
## # A tibble: 5 x 2
##   Country EconClass
##   <chr>    <fct>
## 1 Belgium Low growth
## 2 Bulgaria High growth
## 3 Czechia Middle growth
## 4 Denmark Middle growth
## 5 Germany Low growth
```

We use this data to clearly show the unemployment rates in different types of countries with varying levels

of growth, in order to identify a possible relationship between them. We use a boxplot for this, so the three different groups are clearly arranged side by side and easy to compare.

Variable 2: Cumulative GDP Growth

We also create variables to assess cumulative GDP growth before and after the COVID-19 pandemic.

```
GDP_df <- GDP_df %>%
  rowwise() %>%
  mutate(
    GDP_before_covid = round((prod(1 + c_across(`2015`:`2019`) / 100, na.rm = TRUE) - 1) * 100, 2))

GDP_df <- GDP_df %>%
  rowwise() %>%
  mutate(GDP_after_covid = round((prod(1 + c_across(`2020`:`2024`) / 100, na.rm = TRUE) - 1) * 100, 2))
  ungroup()
```

After creating these two variables, we created a new data frame called `growth_df`. In this data frame, we calculated the average GDP growth per year across all countries. We then used this data to create two plots: EU GDP Growth vs. Average Unemployment and Cumulative EU GDP Growth.

```
gdp_growth <- GDP_df %>%
  select(all_of(as.character(2013:2024)))
avg_growth_per_year <- colMeans(gdp_growth, na.rm = TRUE) / 100

growth_df <- data.frame(
  Year = 2013:2024,
  GrowthRate = avg_growth_per_year
)

growth_df <- growth_df %>%
  mutate(
    TotalGDP = cumprod(1 + GrowthRate) * 100
  )
```

We used the `GDP_df` data, which contains annual GDP growth rates from 2015 to 2024. Using `c_across()` and `prod()` inside `mutate()`, we calculate the compounded GDP growth rate for two periods:

- Pre-COVID: 2015–2019
- Post-COVID: 2020–2024

This transformation is informative because it helps assess how economic performance shifted due to the pandemic. By comparing these two variables, we can identify which countries experienced economic resilience and which struggled to recover.

```
head(GDP_df, 5)[ , c(1, 14, 15, 16)]

## # A tibble: 5 x 4
##   Country GDP_growth_total GDP_before_covid GDP_after_covid
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 Belgium         17.3             8.79           7.78
## 2 Bulgaria        32.3            16.4           13.7
## 3 Czechia         23.4            20.7            2.26
## 4 Denmark         28.0            12.5           13.8
## 5 Germany          9.47             9.1            0.34
```



```
head(growth_df)[,c(2,3)]
```

```
##      GrowthRate TotalGDP
## 2013 0.01174359 101.1744
## 2014 0.02005128 103.2030
## 2015 0.03246154 106.5532
## 2016 0.02794872 109.5312
## 2017 0.03810256 113.7046
## 2018 0.03400000 117.5706
```

These variables provide a basis for analyzing and comparing GDP performance across countries before and after the pandemic. This allows us to assess the relative economic resilience and recovery of different European countries, and identify which economies grew more strongly or faced longer-term setbacks due to COVID-19.

We visualize the average cumulative GDP growth across EU countries from 2013 to 2024 in a line chart, setting the base year to 100. The red dashed line at 2020 highlights the start of the COVID-19 pandemic, allowing us to visually compare pre- and post-pandemic trends.

Variable 3:

```
gdp_growth <- GDP_df %>%
  select(all_of(as.character(2015:2024)))

avg_growth_per_year <- colMeans(gdp_growth, na.rm = TRUE) / 100
```

Variable 3 – EU-wide Average Growth & Unemployment

To get a single time series for the whole EU, we first compute the **average annual GDP growth** by taking the mean of each year's country-level growth rates (2015–2024) and dividing by 100 to turn percentages into proportions. We store that in `avg_growth_per_year`. We then build `growth_df`, a data frame with:

- **Year** – from 2015 to 2024
- **GrowthRate** – the mean annual GDP growth for the EU
- **TotalGDP** – a cumulative GDP index (base 100 in 2015) calculated by compounding each year's GrowthRate with `cumprod(1 + GrowthRate)`.

Next, we summarize our unemployment data into `unemp_df` by:

- Taking the row-wise mean of each year's unemployment figures across all countries,
- Pivoting into long format so we have one row per year with the **average EU-wide unemployment** level.

These two new variables give us a concise picture of the EU's macroeconomic trajectory and its labor market over time, and they form the basis for our event-analysis plots (e.g. overlaying GDP and unemployment trends).

Variable 4:

```

GDP_df <- GDP_df %>%
  rowwise() %>%
  mutate(
    GDP_before_covid = round((prod(1 + c_across(`2015`:`2019`) / 100, na.rm = TRUE) - 1) * 100, 2),
    GDP_after_covid = round((prod(1 + c_across(`2020`:`2024`) / 100, na.rm = TRUE) - 1) * 100, 2)
  ) %>%
  ungroup()

gdp_growth <- GDP_df %>%
  select(all_of(as.character(2013:2024)))
avg_growth_per_year <- colMeans(gdp_growth, na.rm = TRUE) / 100 # omzetting naar factor

growth_df <- data.frame(
  Year = 2013:2024,
  GrowthRate = avg_growth_per_year
)
growth_df <- growth_df %>%
  mutate(
    TotalGDP = cumprod(1 + GrowthRate) * 100 # basisjaar = 100
  )

colnames(Total_Work_population)[1] <- "Country"
colnames(Unemploymentlang)[1] <- "Country"

merged_df <- inner_join(Total_Work_population, Unemploymentlang, by = "Country")

years <- 2015:2024

num_x <- paste0(years, ".x")
num_y <- paste0(years, ".y")

merged_df[ num_x ] <- lapply(merged_df[ num_x ], as.numeric)
merged_df[ num_y ] <- lapply(merged_df[ num_y ], as.numeric)

unemployment_percent_df <- merged_df["Country"]

for (year in years) {
  pop_col <- paste0(year, ".x") # bv. "2015.x"
  unemp_col <- paste0(year, ".y") # bv. "2015.y"
  perc_col <- paste0(year, "_percent") # bv. "2015_percent"

  unemployment_percent_df[[perc_col]] <-
    merged_df[[unemp_col]] / merged_df[[pop_col]] * 100
}

```

```
head(unemployment_percent_df, n = 3)
```

```

## # A tibble: 3 x 11
##   Country '2015_percent' '2016_percent' '2017_percent' '2018_percent'
##   <chr>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 Belgium      5.07      4.66      4.23      3.57
## 2 Bulgaria      5.48      4.49      3.79      3.21

```

```
## 3 Czechia          3.27          2.59          1.90          1.49
## # i 6 more variables: '2019_percent' <dbl>, '2020_percent' <dbl>,
## #   '2021_percent' <dbl>, '2022_percent' <dbl>, '2023_percent' <dbl>,
## #   '2024_percent' <dbl>
```

To calculate the unemployment percentage per country, two datasets were used: the Total_work_population data that indicates the working population of a country and the dataset Unemploymentlang, which gives the number of unemployed people in a country. The reason these two datasets were combined is that the absolute number of unemployed people in a country means nothing if you don't compare it to the total population of that country. By dividing the number of unemployed per country by the population, we got a completely new dataset, and it became possible to compare countries fairly regardless of whether the country has a large or small population. With this new variable, it is possible to analyze how the unemployment percentage changes over time per country. Here is a short description of important functions used for the new variables:

-inner_join to merge the datasets

-as.numeric to correctly convert numbers for the calculation

-for loop to calculate the unemployment percentage per year and store it in columns

colnames & rownames to adjust column names and reset row numbers

Variable 5

```
unemployment_percent_df$avg_unemp_percent <-
  round(
    rowMeans(
      unemployment_percent_df[, grep("_percent$", names(unemployment_percent_df))],
      na.rm = TRUE
    ),
  )
```

```
head(unemployment_percent_df[, c(1, 12)], n = 3)
```

```
## # A tibble: 3 x 2
##   Country avg_unemp_percent
##   <chr>      <dbl>
## 1 Belgium      4
## 2 Bulgaria      3
## 3 Czechia      2
```

After the unemployment percentage variable was created, we also wanted to know the average percentage of unemployment over the years. That's why we added a new column to the unemployment percentage dataset with the average unemployment percentage. This gives new insights, because instead of looking at each year separately, it is now visible at a glance how high or low the unemployment is in a country. This makes it easier to compare countries and, for example, display it on a map. Here is a short description of important functions used for the new variables:

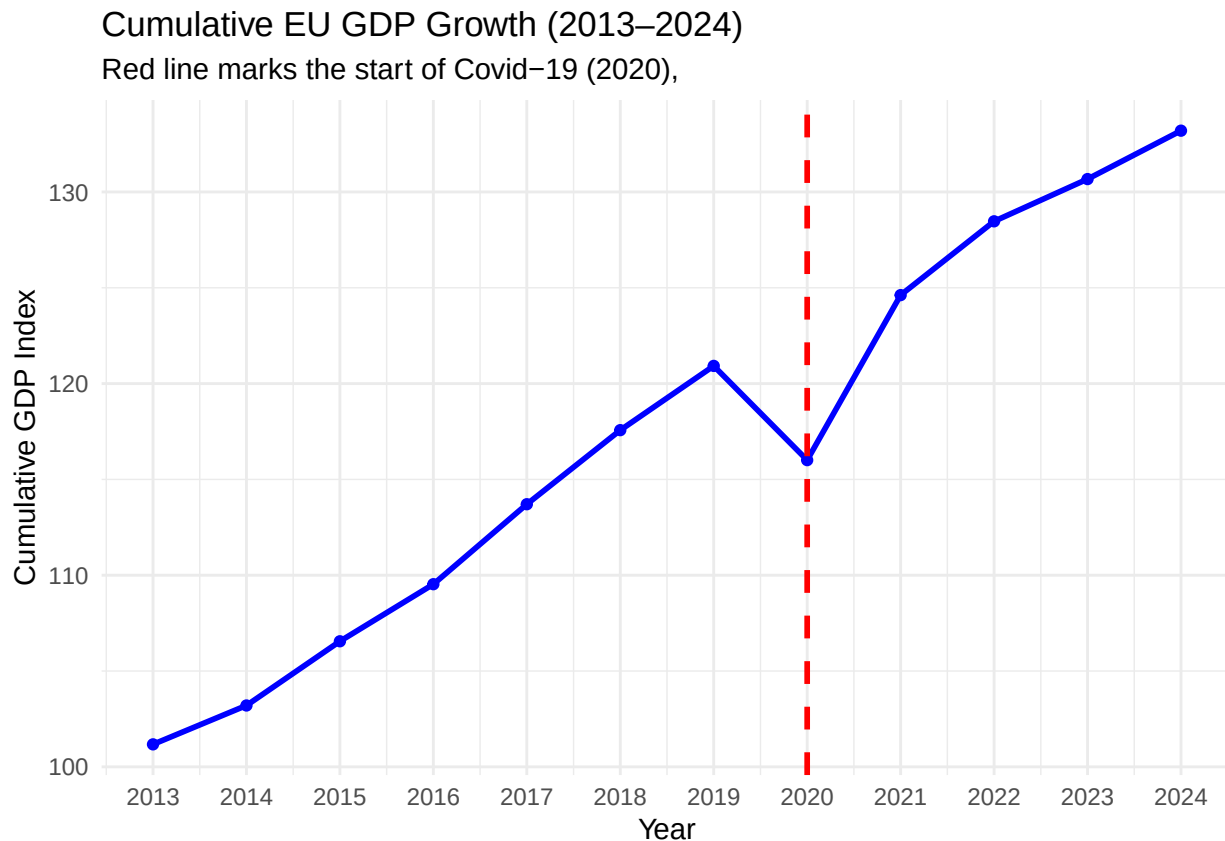
-rowMeans to calculate the average unemployment percentage per country over the entire period

-grep to select the correct columns

-round to neatly round the result to one decimal place

3.3 Visualize temporal variation

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use 'linewidth' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

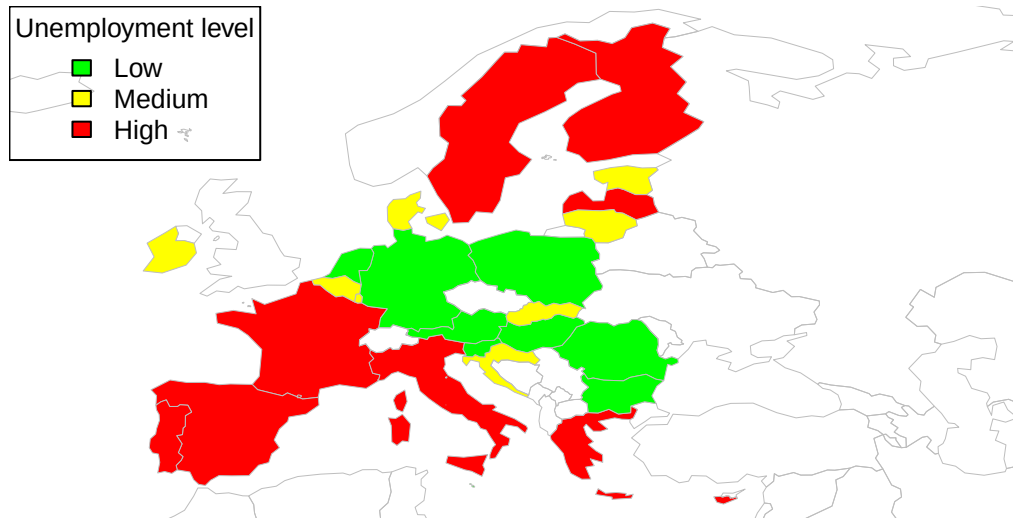


This graph above shows the economic growth called GDP of the European Union over the period 2013 to 2024. As can be seen on the graph, the x-axis shows the years 2013 to 2024, and the y-axis shows GDP growth. And the pink line represents the corona pandemic of 2020. From this graph, you can clearly see that before 2020 there was increasing growth of the GDP, and that around 2020 there was a sharp decline in GDP. In the years after that, the GDP recovered and continued to grow. From this, it can be concluded that the 2020 corona pandemic had a big impact on the economy. And this graph fits well with this project because it shows how the economy can be influenced.

3.4 Visualize spatial variation

```
## 26 codes from your data successfully matched countries in the map  
## 1 codes from your data failed to match with a country code in the map  
## 217 codes from the map weren't represented in your data
```

Average Unemployment in Europe (2015–2024)



To analyze how unemployment varies across Europe, we created a spatial map using the `rworldmap` package and calculated average unemployment percentages from 2015–2024. This visualization is directly tied to our social problem: understanding unemployment in the context of EU economic development. While earlier plots showed trends over time, this spatial map reveals important geographic differences across countries. We calculated unemployment percentages using the total work population. This adjustment allowed for fair comparison between countries of different sizes. Then, we averaged the values over ten years. The map highlights a clear North–South divide in Europe:

- Southern Europe (e.g., Spain, Greece, Italy) shows high unemployment
- Northern and Western Europe (e.g., Germany, Netherlands, Denmark) show low unemployment
- Eastern Europe shows mixed results

This indicates that location is a key factor in unemployment and that regional economic conditions, not just national policies, need to be considered.

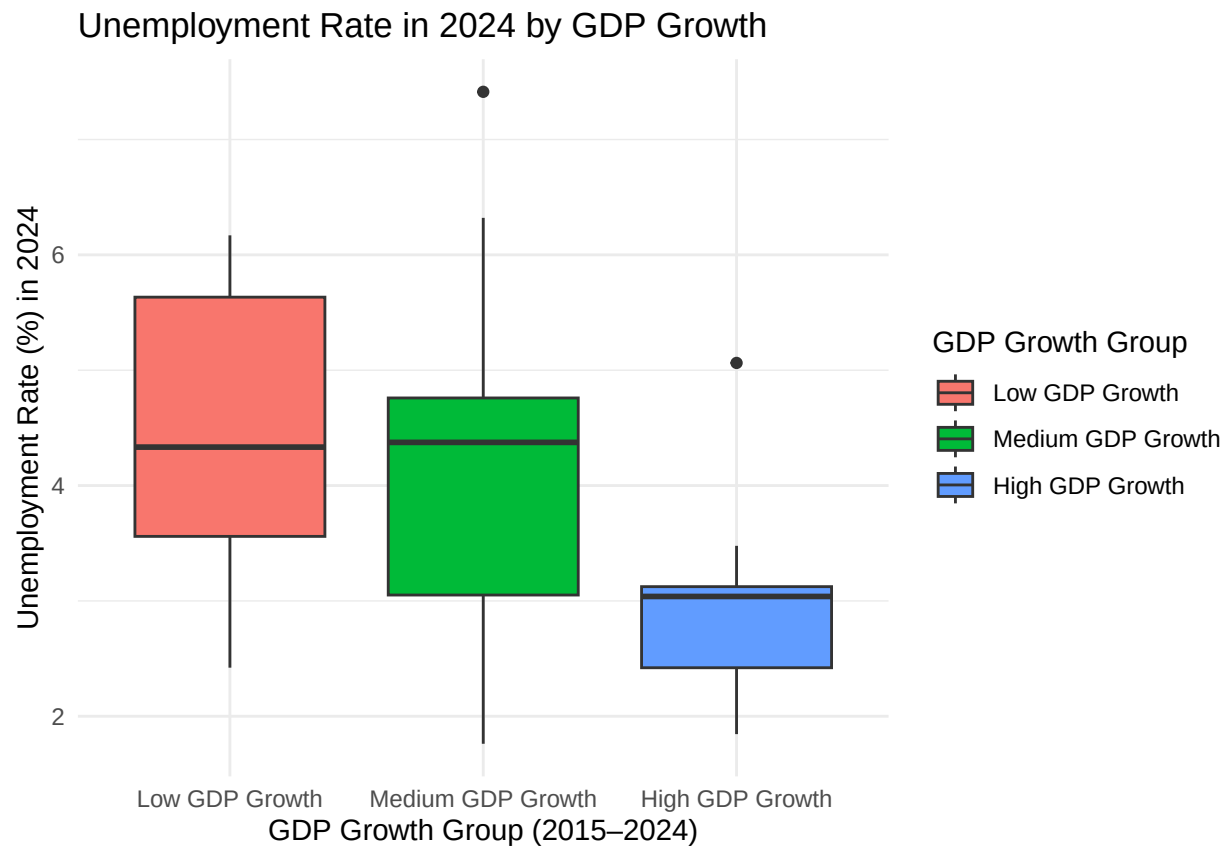
We used `mapRegion = "Europe"` to focus on relevant countries and maintain readability, ensuring that small countries are visible. Although `rworldmap` was used, future improvements could include more detailed shapefiles using the `sf` package for higher resolution.

In summary, this spatial map complements our temporal analysis by showing where unemployment is most severe, reinforcing the need for geographically targeted policies within the EU. The visualization strengthens our understanding of how economic inequality plays out not just over time, but across space, aligning directly with our research topic.

3.5 Visualize sub-population variation

To explore how unemployment varies across economic performance levels, we classified European countries into three groups: Low, Middle, and High growth, based on their total cumulative GDP growth from 2015 to 2024 using data from Eurostat (2025). This grouping allows us to analyze how different economic contexts relate to labor market outcomes.

We merged unemployment and GDP data, then plotted 2024 unemployment rates across these economic classes:



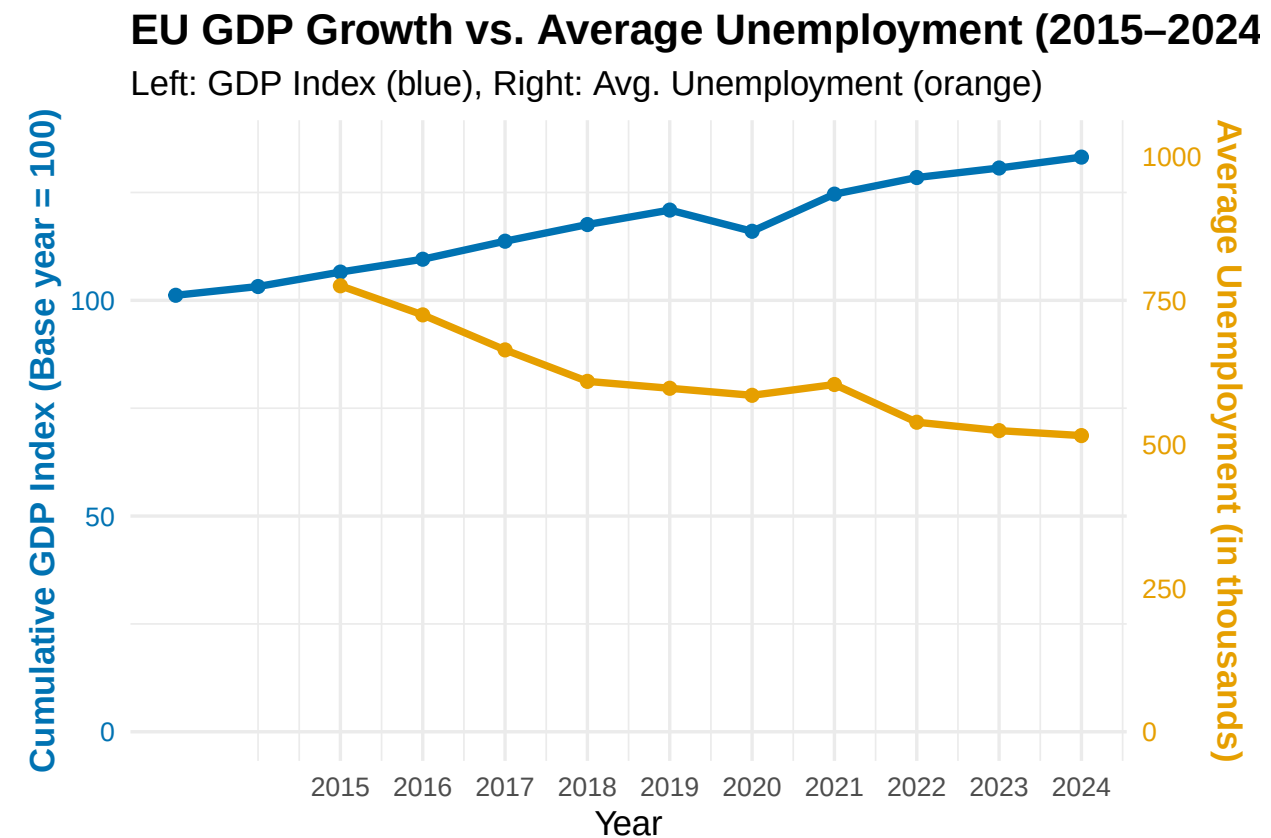
The plot shows that countries in the High growth group tend to have lower and less variable unemployment rates, while Low and Middle growth countries show higher medians and wider spreads. This supports our project’s core question by highlighting how macroeconomic growth correlates with labor market outcomes.

3.6 Event analysis

In this graph, we show 2 variables, first, total GDP growth in Europe, this line is blue. The orange line is about average unemployment. We have chosen that the blue and orange lines both start in the same point to create a clear picture. Therefore, we have taken 2015 as the base year for the Cumulative GDP Index, and adjusted the Y-axis of average unemployment accordingly. To overlay the unemployment series with the GDP index on one chart, we first divide the average-unemployment values by 7.5 so they sit on the same scale as the GDP index. Then we use a secondary y-axis that multiplies those scaled values by 7.5, restoring the true unemployment figures for labeling.

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## ('geom_line()').
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## ('geom_point()').
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This plot is relevant to our social problem, unemployment in Europe. By showing cumulative GDP growth averaged out in Europe and aligning this with the average unemployment, we can come to the conclusion that in periods with higher GDP growth, there is also lower unemployment. On the other hand, you can also see that around the corona pandemic in 2020-2021, there is a drop in GDP and unemployment rises. This again shows the link between GDP and unemployment, and that investment in new green technologies, for example, can help in rising GDP and lower unemployment.

4.1 Discuss your findings

We have used three different datasets to create four charts. With the use of the data found, we made charts which we could use to visualize our findings. We have made the following charts:

- Chart 1: Cumulative EU GDP growth (2015-2024)
- Chart 2: Unemployment rate in 2024 by GDP Growth
- Chart 3: EU GDP growth vs average unemployment (2015-2024)
- Chart 4: Average unemployment in Europe (2015-2024)

Cumulative EU GDP growth (2015-2024). This chart shows us the cumulative GDP index. On the X-axis, it shows the year starting from 2015 until 2024. On the Y-axis we have the Cumulative GDP index. For this chart we decided to use the line chart since it offers a nice and accurate visual of the annual change of the GDP index. And we also added a vertical red line in 2020 to give a visual representation of the start of COVID-19. Which clarifies when COVID-19 starts and the effect it has before and after.

Unemployment rate in 2024 by GDP Growth. This chart gives us insight into the percentage rate of unemployment in 2024. We used the boxplot because it shows distributions of numeric data values, especially when you want to compare them between multiple groups. In this case our different groups are sorted by GDP growth. Dividing it into three groups: low growth, middle growth, and high growth.

EU GDP growth vs average unemployment (2015-2024). This chart compares two economic indicators over a period of 10 years (2015-2024). Noticeably, it contains two y-axes. The left y-axis (blue) is the cumulative GDP Index with 100 as the base year and the right y-axis (orange) is the average unemployment in thousands. With the help of this chart, we can visually analyse the relationship between GDP growth and the average unemployment.

Average unemployment in Europe (2015-2024). The last chart is a map of Europe. It illustrates the average unemployment rate across various European countries for the years 2015-2024. Each country is colour-coded according to its unemployment rate. Green means low unemployment, yellow means medium unemployment and red indicates high unemployment. This way unemployment rates between countries can be easily compared.

Based on the charts and datasets provided, we can draw up the following findings. In the chart of *EU GDP growth vs average unemployment (2015-2024)* we can derive that the EU experienced a steady economic growth from 2015-2019, a small recession in 2020 and a strong recovery afterwards. Unemployment consistently decreased over the period, showing only minor disruptions during 2020. Because of this we can conclude that when the GDP increases, unemployment decreases. Looking at the map of average unemployment in Europe, we can gather that the majority of central and some of the northern countries have an overall low unemployment rate in comparison to southern and some western countries. But also, when you look at chart 2 you can conclude that the unemployment rate will be higher when the GDP growth group is lower.

Part 5 - Reproducibility

5.1 Github repository link

<https://github.com/benjaminmutlu/PfE>

5.2 Reference list

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